

Discrete configurations of points and lines

- 1 Let 100 points be given inside a unit circle. Prove that one can find two of these points that are on a distance not exceeding $\frac{2}{5}$.
- 2 Finitely many segments are given in the plane, with the total length less than $\sqrt{2}$. Prove that there exists an infinite unit square grid that meets none of these segments.
- 3 Let S be a finite set of points in the plane, not all of which lie on a single line. Prove that there is a line containing exactly two of these points.
- 4 Prove that among (a) any four points in a unit circle; (b) any five points in a unit sphere there are two on a distance not exceeding $\sqrt{2}$.
- 5 For which positive integers n is there a closed polygonal line consisting of n unit segments such that no two vertices of the polygonal line are on a distance greater than 1?
- 6 Let n^2 points be given inside a unit square. Prove that all these points can be connected by a polygonal line of length less than $2n + 1$.
- 7 In the plane are given $2n$ distinct points, n of which are red and n are blue. Prove that it is always possible to draw n pairwise disjoint segments, each of which connects a red and a blue point.
- 8 An L -line is a polygonal line in the (cartesian) plane consisting of a vertical and a horizontal segment. There are n red and n blue points in the plane, with distinct x -coordinates and distinct y -coordinates. Prove that these points can be connected by using n disjoint L -lines, each of which has one red and one blue endpoint.
- 9 There are n points in the plane, no three of which are collinear, and a line ℓ passing through one of these points, call it P . The line starts rotating clockwise around P until it meets another point for the first time, say Q . Then it starts rotating clockwise around Q until it meets some other point for the first time, call it R . The process continues indefinitely. Prove that there exists an initial position of the line such that the line eventually reaches every point.
- 10 Prove that among any five distinct points in the plane, no three of which are collinear, there are four that are vertices of a convex quadrilateral.
- 11 Some n points in the plane, no three of which are collinear, are painted red. Prove that one can always find $2n - 5$ points and paint them black such that every triangle with the vertices at the red points contains at least one black point in its interior.
- 12 Ten distinct points, denoted $A_1, A_2, \dots, A_{10}$, are given in the plane. For every oriented line ℓ , we denote the orthogonal projection of point A_i by number i . These projections, assuming they are distinct, yield a permutation of numbers $1, 2, \dots, 10$. At most how many different permutations can we obtain in this way?
- 13 Let S be an n -point set in the plane, where $n \in \mathbb{N}$. An *eligible* set is any nonempty set of the form $S \cap D$, where D is a closed disk in the plane. At least how many eligible sets can there be?
- 14 We are given 999 points in the plane, no three collinear, of which 499 are red and 500 are blue. Drawing lines will divide the plane into regions. We should draw k lines, passing through none of these points, so that no region contains points of different colors. What is the smallest k for which this is always possible?
- 15 A set of $2n + 1$ points in the plane is given so that no three points are collinear and no four are concyclic. A circle is *balanced* if it passes through three given points and has $n - 1$ points inside and $n - 1$ points outside. Prove that there are exactly n^2 balanced circles.

- 16** Let $n > 2$ be an integer. A map f assigns each point in the plane a real number. Suppose that, for every regular n -gon $A_1 A_2 \dots A_n$, the sum $f(A_1) + f(A_2) + \dots + f(A_n)$ is zero. Prove that f is the zero map.
- 17** In the plane are given n points. Any three points form a non-degenerate triangle of area at most 1. Prove that not more than n of these triangles can have area equal to 1.
- 18** Prove that any n lines in the plane, no two of which are parallel and no three concurrent, divide the plane into exactly $2n$ unbounded and $\frac{(n-1)(n-2)}{2}$ bounded regions.
- 19** In the plane are given n *directed* lines, no two of which are parallel and no three concurrent. A region is *orientable* if one can travel along its boundary by following the directions of the respective lines. Prove that there are at most $\frac{n^2+n}{3}$ orientable regions.
- 20** We are given n lines in the plane, no two of which are parallel and no three concurrent, and a point O not lying on any line. Denote the set of intersection points of these lines by S . Given an integer $0 \leq k \leq n-2$, we paint red every point $P \in S$ with the property that the segment OP (without the endpoints) intersects at most k lines. Prove that there are at least $\frac{(k+1)(k+2)}{2}$ red points.
- 21** Given an arrangement of $n \geq 3$ lines in a general position, find the smallest and largest possible number of unbounded angular faces (that is, regions bounded by only two rays) they can form.
- 22** A plane is divided into regions by n lines in a general position (no two parallel, no three concurrent). Show that it is possible to write in each region an integer not exceeding n in absolute value in such a way that, for each line, the sum of numbers on either side of this line is zero.
- 23** Prove that every arrangement of n lines, with no two parallel and no three concurrent, forms at least $n-2$ and at most $\frac{n(n-2)}{3}$ triangles. Is $n-2$ attainable?
- 24** A set of n lines in a general position dissect the plane into regions. Let ℓ be one of these n lines; then $2n$ of these regions have an edge on ℓ . Prove that the sum of the numbers of edges of these $2n$ regions is less than $10n$.
- 25** A set of n lines are given in the plane, no two of which are parallel and no three concurrent. Prove that there is a line in the given set, such that each of the two open half-planes determined by this line contains at least $\frac{1}{10}(n-2)^2$ intersection points of these n lines.